Black–Scholes from Replication

Itô practice, self-financing strategies, the pricing PDE, verification, and Greeks

Lecture 5

Today one idea will connect calculus, pricing, and hedging

1. Use Itô's formula on three new objects.
2. Define a self-financing trading strategy precisely.
3. Replicate $g(S_T)$ and derive the Black–Scholes PDE.
4. Solve the PDE through time reversal, discounting, and $\log S$.
5. Verify the candidate price and recover the hedge.
6. Specialize to a call and compute its Greeks.

Central message

The price is the initial wealth of a self-financing strategy that reproduces the payoff in every state of the model.

Three New Itô Calculations

The formula is a chain rule plus quadratic variation

If

$$\mathrm{d}X_t = b_t\,\mathrm{d}t + a_t\,\mathrm{d}W_t, \qquad f \in C^{1,2},$$

then

$$\boxed{\mathrm{d}f(t, X_t) = \left(f_t + b_t f_x + \frac{1}{2}a_t^2 f_{xx}\right)\mathrm{d}t + a_t f_x\,\mathrm{d}W_t.}$$

Calculation habit

Compute f_t, f_x, f_{xx} first. Substitute the process only afterward.

Logical habit

Zero drift first gives a local martingale. Check integrability before claiming a true martingale.

A two-dimensional Brownian motion gives drift 2

Let $W = (W^1, W^2)$ be a standard two-dimensional Brownian motion and

$$R_t = \|W_t\|^2 = (W_t^1)^2 + (W_t^2)^2.$$

This is the squared radius, or squared Bessel process of dimension 2.

YOUR TURN Compute $\mathrm{d}R_t$. Which deterministic term makes the result a martingale?

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Applying Itô to each squared coordinate gives

$$\boxed{\mathrm{d}R_t = 2W_t^\top \,\mathrm{d}W_t + 2\,\mathrm{d}t,} \qquad \boxed{R_t - 2t = 2\int_0^t W_s^\top \,\mathrm{d}W_s.}$$

On bounded horizons the integrand is square-integrable, so $R_t - 2t$ is a true martingale.

In d dimensions, the drift counts independent directions

Now let W be a standard d-dimensional Brownian motion. Then

$$\mathbb{E}\|W_t\|^2 = dt, \qquad [\,\|W\|^2\,]_t = 4\int_0^t \|W_s\|^2\,\mathrm{d}s.$$

Interpretation

Each Brownian coordinate contributes one unit of quadratic variation per unit of time. Dimension appears as drift after a nonlinear transformation.

SIMPLIFICATION The coordinates are independent and normalized: $\mathrm{d}[W^i, W^j]_t = \delta_{ij}\,\mathrm{d}t$. With covariance matrix Σ, the correction becomes $\mathrm{tr}(\Sigma)\,\mathrm{d}t$.

Example 2 combines a finite-variation state with Brownian motion

Define integrated Brownian motion

$$A_t = \int_0^t W_s \, \mathrm{d}s, \qquad Y_t = W_t A_t.$$

YOUR TURN Use the product rule to find $\mathrm{d}Y_t$, then compute $\mathbb{E}[Y_t]$.

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Since $\mathrm{d}A_t = W_t \, \mathrm{d}t$ and A has finite variation, $\mathrm{d}[W, A]_t = 0$. Therefore

$$\boxed{\mathrm{d}(W_t A_t) = A_t \, \mathrm{d}W_t + W_t^2 \, \mathrm{d}t.}$$

Taking expectations,

$$\boxed{\mathbb{E}[W_t A_t] = \int_0^t \mathbb{E}[W_s^2] \, \mathrm{d}s = \frac{t^2}{2}.}$$

The same answer is a covariance computed two ways

Because both variables are centered,

$$\mathbb{E}[W_t A_t] = \mathrm{Cov}\left(W_t, \int_0^t W_s \, \mathrm{d}s\right).$$

YOUR TURN Verify the answer directly from $\mathrm{Cov}(W_t, W_s) = \min(t, s)$.

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$$\mathrm{Cov}\left(W_t, \int_0^t W_s \, \mathrm{d}s\right) = \int_0^t \mathrm{Cov}(W_t, W_s) \, \mathrm{d}s = \int_0^t s \, \mathrm{d}s = \frac{t^2}{2}.$$

Why this example matters

Itô calculus handles path-dependent state variables after we enlarge the state from W_t to (W_t, A_t).

Example 3 turns a future event into a martingale

Fix $T > 0$. For $t < T$, consider

$$u(t, x) = \mathrm{N}\left(\frac{x}{\sqrt{T - t}} \right),$$

where N and φ are the standard normal cdf and density.

YOUR TURN Compute u_t, u_x, u_{xx}. Does $u(t, W_t)$ have drift?

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With $\tau = T - t$ and $z = x/\sqrt{\tau}$,

$$u_x = \frac{\varphi(z)}{\sqrt{\tau}}, \qquad u_{xx} = -\frac{x\varphi(z)}{\tau^{3/2}}, \qquad u_t = \frac{x\varphi(z)}{2\tau^{3/2}}.$$

The backward heat equation removes the drift

The derivatives satisfy

$$\boxed{u_t + \frac{1}{2} u_{xx} = 0.}$$

Itô's formula therefore gives

$$\boxed{\mathrm{d}u(t, W_t) = \frac{\varphi(W_t/\sqrt{T-t})}{\sqrt{T-t}}\, \mathrm{d}W_t.}$$

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Moreover,

$$u(t, W_t) = \mathbb{P}(W_T > 0 \mid \mathcal{F}_t),$$

because $W_T = W_t + \sqrt{T-t}\, Z$, with $Z \sim N(0,1)$ independent of $\mathcal{F}_t$.

Preview

A terminal payoff, a backward PDE, and a conditional expectation are three views of the same object. Black–Scholes will add discounting and replication.

Replication Creates the Pricing Equation

Black–Scholes starts with a model and a trading technology

MODEL ASSUMPTION One stock follows

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \quad \sigma > 0.$$

One bank account satisfies

$$\mathrm{d}B_t = rB_t\,\mathrm{d}t, \quad B_t = e^{rt}.$$

MARKET ASSUMPTION Trading is continuous and frictionless. Stock and bank account can be bought or shorted, and borrowing/lending occurs at the same rate r.

SIMPLIFICATION For now, r, μ, σ are constant, the stock pays no dividend, and its paths are continuous.

The assumptions have different logical roles

Type	Assumption	Where it enters
Model	GBM, constant σ, no jumps	Itô dynamics and PDE coefficients
Market	No arbitrage	A locally riskless position must earn r
Market	Continuous frictionless trading	Delta can be adjusted continuously
Market	One Brownian risk, one stock	Every smooth European claim can be hedged
Simplification	Constant r, σ, no dividends	Closed-form $d_{\pm}$ formula
Regularity	$p \in C^{1,2}$ before maturity	Classical Itô formula applies

Do not merge these claims

A realistic price model need not create a complete market; a no-arbitrage market need not admit a closed-form formula.

A European claim fixes only the terminal value

The claim pays

$$H = g(S_T)$$

at maturity T. We seek a deterministic Markov price

$$V_t = p(t, S_t), \qquad \boxed{p(T, s) = g(s).}$$

MODEL ASSUMPTION The Markov representation assumes the current stock price is a sufficient state variable for this claim.

SIMPLIFICATION We initially assume $p \in C^{1,2}([0, T) \times (0, \infty))$. Vanilla payoffs have a kink at maturity, but the price becomes smooth for $t < T$.

A trading strategy separates holdings from gains

At time t, hold

$$\Delta_t \text{ shares of stock}, \qquad \beta_t \text{ units of the bank account.}$$

The portfolio value is

$$\boxed{X_t = \Delta_t S_t + \beta_t B_t.}$$

MARKET ASSUMPTION Δ and β must be predictable: the position used over the next instant is chosen using currently available information.

Replication objective

Choose X_0, Δ, β so that $X_T = g(S_T)$ in every state allowed by the model.

Self-financing is a budget constraint on rebalancing

change in wealth = gain on the assets + external cash flow.

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$$\text{change in wealth} = \text{gain on the assets} + \text{external cash flow}.$$

Self-financing

After the initial investment, the external cash flow is zero.

Allowed

Δ and β may change through time as information arrives.

Not allowed

Rebalancing cannot create wealth or require a new deposit.

We first write this budget constraint in discrete time, where every cash flow is visible, and then pass to continuous time.

In discrete time, rebalancing preserves current wealth

At date t_n, value the old and new holdings at the *same* prices:

$$X_n^- = \Delta_{n-1} S_n + \beta_{n-1} B_n, \qquad X_n = \Delta_n S_n + \beta_n B_n.$$

YOUR TURN If no cash enters or leaves at t_n, what equality must hold?

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$$X_n = X_n^- \iff \boxed{S_n(\Delta_n - \Delta_{n-1}) + B_n(\beta_n - \beta_{n-1}) = 0.}$$

Interpretation

Buying additional shares is financed by reducing cash; selling shares increases cash. Rebalancing only changes the composition of wealth.

In discrete time, gains come only from asset-price moves

After rebalancing at t_n, keep (Δ_n, β_n) over $(t_n, t_{n+1}]$. Just before the next rebalance,

$$X_{n+1}^{-} = \Delta_n S_{n+1} + \beta_n B_{n+1}.$$

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$$\boxed{X_{n+1}^{-} - X_n = \Delta_n(S_{n+1} - S_n) + \beta_n(B_{n+1} - B_n).}$$

Key distinction

Rebalancing changes holdings but not wealth. Asset-price changes generate gains and losses.

Continuous self-financing is the limit of discrete gains

On a trading grid, accumulated wealth satisfies

$$X_t - X_0 = \sum_n \Delta_n (S_{n+1} - S_n) + \sum_n \beta_n (B_{n+1} - B_n).$$

As the mesh tends to zero, the predictable left-point sums become stochastic integrals:

$$\boxed{X_t = X_0 + \int_0^t \Delta_u \, \mathrm{d}S_u + \int_0^t \beta_u \, \mathrm{d}B_u,} \qquad \boxed{\mathrm{d}X_t = \Delta_t \, \mathrm{d}S_t + \beta_t \, \mathrm{d}B_t.}$$

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Together with $X_t = \Delta_t S_t + \beta_t B_t$, the product rule yields the equivalent rebalancing constraint

$$\boxed{S_t \, \mathrm{d}\Delta_t + \, \mathrm{d}[\Delta, S]_t + B_t \, \mathrm{d}\beta_t = 0.}$$

First calculation: expand the option dynamics

$$V_t = p(t, S_t), \qquad \mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t.$$

YOUR TURN Use Itô's formula. Separate the drift and diffusion coefficients.

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$$\boxed{\begin{aligned}\mathrm{d}V_t = &\left(p_t + \mu S_t p_s + \frac{1}{2}\sigma^2 S_t^2 p_{ss}\right) \mathrm{d}t \\ &+ \sigma S_t p_s \,\mathrm{d}W_t.\end{aligned}}$$

All derivatives are evaluated at (t, S_t).

The bank-account gain comes from the cash balance

The portfolio value is

$$X_t = \Delta_t S_t + \beta_t B_t \quad \Longrightarrow \quad \boxed{\beta_t B_t = X_t - \Delta_t S_t.}$$

Since $\mathrm{d}B_t = rB_t\,\mathrm{d}t$, the bank-account gain is

$$\boxed{\beta_t\,\mathrm{d}B_t = r\beta_t B_t\,\mathrm{d}t = r(X_t - \Delta_t S_t)\,\mathrm{d}t.}$$

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YOUR TURN Now insert $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$.

$$\boxed{\mathrm{d}X_t = [rX_t + (\mu - r)S_t\Delta_t]\,\mathrm{d}t + \sigma S_t\Delta_t\,\mathrm{d}W_t.}$$

MARKET ASSUMPTION Here r applies to both positive and negative bank balances.

Matching random shocks determines the hedge

Replication requires

$$X_t = V_t = p(t, S_t) \quad \text{and} \quad \mathrm{d}X_t = \mathrm{d}V_t.$$

YOUR TURN Match the $\mathrm{d}W_t$ coefficients. What must Δ_t be?

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YOUR TURN Match the $\mathrm{d}W_t$ coefficients. What must Δ_t be?

$$\sigma S_t \Delta_t = \sigma S_t p_s(t, S_t).$$

Since $S_t > 0$ and $\sigma > 0$,

$$\boxed{\Delta_t = p_s(t, S_t).}$$

Economic meaning

Delta is not merely a sensitivity. In this complete diffusion model, it is the stock holding in the replicating strategy.

Matching drifts produces one final cancellation

Set $X_t = p(t, S_t)$ and $\Delta_t = p_s(t, S_t)$.

YOUR TURN Match the drift coefficients and simplify. Which parameter disappears?

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YOUR TURN Match the drift coefficients and simplify. Which parameter disappears?

$$rp + (\mu - r)sp_s = p_t + \mu sp_s + \frac{1}{2}\sigma^2 s^2 p_{ss}.$$

The terms containing the physical drift cancel:

$$\boxed{p_t + rsp_s + \frac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0,} \qquad \boxed{p(T, s) = g(s).}$$

The Black–Scholes PDE is a replication equation

$$\begin{cases} p_t(t,s) + rs\, p_s(t,s) + \frac{1}{2}\sigma^2 s^2 p_{ss}(t,s) - rp(t,s) = 0, \\ p(T,s) = g(s). \end{cases}$$

What created each term?

- p_t: passage of calendar time;
- rsp_s: financing the stock position;
- $\frac{1}{2}\sigma^2 s^2 p_{ss}$: quadratic variation and gamma;
- $-rp$: financing the total option value.

Why does the expected stock return disappear?

The stock exposure appears twice:

$$\underbrace{\mu s p_s}_{\text{option's stock exposure}} \quad \text{and} \quad \underbrace{\mu s \Delta}_{\text{replicating stock position}} .$$

With $\Delta = p_s$, they are identical and cancel.

Pricing conclusion

In the model, valuation depends on r and σ, not on the estimated physical drift μ.

What this does not say

Real-world probabilities remain essential for forecasting, risk management, and expected returns.

Replication needs no arbitrage and enough traded risk

MARKET ASSUMPTION **No arbitrage.** Two self-financing portfolios with the same terminal payoff must have the same initial value.

MARKET ASSUMPTION **Completeness.** One stock spans the one Brownian source of uncertainty, so the option's diffusion exposure can be matched.

When the argument breaks

With jumps, stochastic volatility driven by another shock, trading constraints, or transaction costs, one stock generally cannot eliminate all risk. The replication price may cease to be unique.

The bank-account holding follows from the value identity

Once $\Delta_t = p_s(t, S_t)$,

$$\beta_t = \frac{p(t, S_t) - S_t p_s(t, S_t)}{B_t}.$$

YOUR TURN At maturity, what does the portfolio hold when $g(s) = (s - K)^+$, away from the strike?

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$$\Delta_T = \begin{cases} 1, & S_T > K, \\ 0, & S_T < K, \end{cases} \qquad \beta_T B_T = \begin{cases} -K, & S_T > K, \\ 0, & S_T < K. \end{cases}$$

The call is replicated by one share financed by K of borrowing when it finishes in the money.

Self-financing is not the same as zero transaction costs

MARKET ASSUMPTION The model has no commissions or bid–ask spread. But buying shares still requires cash: the bank-account holding must adjust to pay for them.

1. **Rebalancing does not create wealth.** Buying shares is paid for from the cash account; this is funding, not a transaction fee.
2. **Matching Delta removes only the Brownian shock.** The portfolio must also satisfy the self-financing wealth equation.
3. **The option need not earn r by itself.** After matching its stock exposure, the whole locally riskless portfolio earns r.
4. **μ is not replaced by r.** It cancels because the option and its replicating stock position have the same exposure.

Solve the PDE by Taking Logs

The terminal-value problem is our starting point

$$p_t + rsp_s + \frac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0, \qquad p(T, s) = g(s).$$

The coefficient s^2 prevents this from being the standard heat equation.

Four transformations

time to maturity $\longrightarrow$ remove discounting $\longrightarrow$ $\log s$ $\longrightarrow$ remove transport.

SIMPLIFICATION Constant r and σ make these transformations explicit. The pricing principle survives more general coefficients, but the closed form usually does not.

Step 1 turns the terminal condition into initial data

Set

$$\tau = T - t, \qquad v(\tau, s) = p(T - \tau, s).$$

YOUR TURN Transform the PDE and its terminal condition.

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Because $p_t = -v_\tau$,

$$\boxed{v_\tau = rsv_s + \frac{1}{2}\sigma^2 s^2 v_{ss} - rv, \qquad v(0, s) = g(s).}$$

Step 2 removes the financing term

Define

$$u(\tau, s) = e^{r\tau} v(\tau, s).$$

YOUR TURN Compute u_τ, u_s, u_{ss} and simplify the PDE.

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$$u_\tau = ru + e^{r\tau} v_\tau, \qquad u_s = e^{r\tau} v_s, \qquad u_{ss} = e^{r\tau} v_{ss}.$$

Thus

$$\boxed{u_\tau = rsu_s + \frac{1}{2}\sigma^2 s^2 u_{ss}, \qquad u(0, s) = g(s).}$$

Step 3 makes multiplicative motion additive

Let

$$x = \log s, \qquad w(\tau, x) = u(\tau, e^x).$$

YOUR TURN Express u_s and u_{ss} using w_x, w_{xx}.

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$$x = \log s, \qquad w(\tau, x) = u(\tau, e^x).$$

YOUR TURN Express u_s and u_{ss} using w_x, w_{xx}.

$$w_x = s u_s, \qquad w_{xx} = s u_s + s^2 u_{ss}.$$

Therefore

$$\boxed{u_s = \frac{w_x}{s}, \qquad u_{ss} = \frac{w_{xx} - w_x}{s^2}.}$$

The log transform leaves heat plus constant transport

Insert the derivatives:

$$\boxed{w_\tau = \frac{1}{2}\sigma^2 w_{xx} + \left(r - \frac{1}{2}\sigma^2\right) w_x, \qquad w(0,x) = g(e^x).}$$

YOUR TURN Why does $r - \sigma^2/2$ look familiar?

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YOUR TURN Why does $r - \sigma^2/2$ look familiar?

It is the drift of $\log S$ when the stock's pricing drift is r:

$$\mathrm{d}\log S_t = \left(r - \frac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Step 4 moves with the transport velocity

Write

$$a = r - \frac{1}{2}\sigma^2, \qquad \varphi(\tau, y) = w(\tau, y - a\tau).$$

YOUR TURN Differentiate φ. Which PDE does it satisfy?

Step 4 moves with the transport velocity

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$$a = r - \frac{1}{2}\sigma^2, \qquad \varphi(\tau, y) = w(\tau, y - a\tau).$$

YOUR TURN Differentiate φ. Which PDE does it satisfy?

$$\varphi_\tau = w_\tau - a w_x, \qquad \varphi_y = w_x, \qquad \varphi_{yy} = w_{xx}.$$

Hence

$$\boxed{\varphi_\tau = \frac{1}{2}\sigma^2 \varphi_{yy}, \qquad \varphi(0, y) = g(e^y).}$$

The heat equation averages terminal values with a Gaussian

If $Z \sim N(0,1)$, then

$$\boxed{\varphi(\tau, y) = \mathbb{E}\left[g\left(e^{y+\sigma\sqrt{\tau}Z}\right)\right].}$$

Equivalently,

$$\varphi(\tau, y) = \int_{\mathbb{R}} g(e^z) \frac{\exp\left(-\frac{(z-y)^2}{2\sigma^2\tau}\right)}{\sqrt{2\pi\sigma^2\tau}} \, \mathrm{d}z.$$

Meaning

Diffusion smooths the terminal payoff by averaging it over future Gaussian log-price shocks.

Undoing the transformations gives the general price

For $\tau = T - t$,

$$p(t,s) = e^{-r\tau}\mathbb{E}\Big[g\Big(s\exp\{(r-\tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\}\Big)\Big], \qquad Z \sim N(0,1).$$

MODEL ASSUMPTION Inside the expectation, the auxiliary stock grows at drift r, not at the physical drift μ.

Scope

This is the general European-payoff formula in the Black–Scholes model. It is a Gaussian integral, not a closed form for every g.

Regularity and growth conditions are doing real work

The formal derivation is valid for a smooth payoff of controlled growth. For common nonsmooth payoffs:

- the terminal condition may have kinks or discontinuities;
- the Gaussian convolution is smooth for every $t < T$;
- the terminal condition is recovered in the appropriate limiting sense.

SIMPLIFICATION We suppress the distinction between classical, weak, and viscosity solutions. For vanilla calls and puts, the pricing formula is well defined and the required derivatives exist before maturity.

Feynman–Kac Verification

Feynman–Kac represents the PDE solution as an expectation

We want to turn the PDE solution p into an expectation. Its spatial drift term is $rs\,p_s$, and its discount term is $-rp$.

YOUR TURN Choose a process whose drift makes Itô's formula reproduce those terms.

Feynman–Kac represents the PDE solution as an expectation

We want to turn the PDE solution p into an expectation. Its spatial drift term is $rs\,p_s$, and its discount term is $-rp$.

YOUR TURN Choose a process whose drift makes Itô's formula reproduce those terms.

For each starting pair (t, s), define S^r on $[t, T]$ by

$$\frac{\mathrm{d}S_u^r}{S_u^r} = r\,\mathrm{d}u + \sigma\,\mathrm{d}W_u, \qquad S_t^r = s.$$

Write $\mathbb{P}_{t,s}^r$ and $\mathbb{E}_{t,s}^r$ for probability and expectation under this process started at s at time t.

SIMPLIFICATION S^r is an auxiliary pricing process, not the physical stock forecast. Its drift r matches the PDE; later, a change of measure gives it a risk-neutral interpretation.

Discounting and the PDE remove the drift

YOUR TURN Apply Itô's formula to $e^{-r(u-t)}p(u, S_u^r)$. Identify its drift and explain why that drift vanishes.

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YOUR TURN Apply Itô's formula to $e^{-r(u-t)}p(u, S_u^r)$. Identify its drift and explain why that drift vanishes.

Itô's formula gives

$$\begin{aligned}\mathrm{d}\big(e^{-r(u-t)}p(u,S_u^r)\big) = e^{-r(u-t)}\Big[&p_t + rS_u^r p_s \\ &+ \frac{1}{2}\sigma^2(S_u^r)^2 p_{ss} - rp\Big]\,\mathrm{d}u \\ &+ e^{-r(u-t)}\sigma S_u^r p_s\,\mathrm{d}W_u.\end{aligned}$$

The drift bracket is exactly the Black–Scholes PDE, evaluated at (u, S_u^r), so it vanishes:

$$\boxed{\mathrm{d}\big(e^{-r(u-t)}p(u,S_u^r)\big) = e^{-r(u-t)}\sigma S_u^r p_s\,\mathrm{d}W_u.}$$

The terminal condition turns the martingale into the price

If the stochastic integral is a true martingale, its expected future value equals its current value. Thus

$$p(t,s) = \mathbb{E}^r_{t,s}\left[e^{-r(T-t)}p(T, S^r_T)\right].$$

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Now use the terminal condition $p(T,x) = g(x)$:

$$\boxed{p(t,s) = \mathbb{E}^r_{t,s}\left[e^{-r(T-t)}g(S^r_T)\right].}$$

Verification condition

For example, it is enough that the displayed stochastic integral has finite expected quadratic variation. Otherwise, localization and uniform integrability are needed.

The same price function gives the hedge in the actual market

Now return to the traded stock S_t, whose physical drift is μ. Given a sufficiently regular PDE solution, define

$$\Delta_t = p_s(t, S_t), \qquad \beta_t = \frac{p(t, S_t) - S_t p_s(t, S_t)}{B_t}.$$

YOUR TURN Check that these holdings are self-financing and match the payoff.

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The PDE rearranges Itô's formula as

$$\mathrm{d}p(t, S_t) = p_s(t, S_t)\,\mathrm{d}S_t + r\big(p(t, S_t) - S_t p_s(t, S_t)\big)\,\mathrm{d}t.$$

Therefore

$$\mathrm{d}p(t, S_t) = \Delta_t\,\mathrm{d}S_t + \beta_t\,\mathrm{d}B_t, \qquad p(T, S_T) = g(S_T).$$

Two roles, one price function

S^r represents the price as an expectation; S is the traded asset used to implement the replicating hedge.

Expectation and replication prove different facts

PDE / replication

Explains why the price is unique, why μ disappears, and why $\Delta = p_s$.

Expectation / verification

Computes the PDE solution and checks terminal matching and discounted martingale dynamics.

self-financing replication $\Longleftrightarrow$ PDE $\Longleftrightarrow$ discounted expectation.

SIMPLIFICATION We built an auxiliary drift-r process directly. A later measure-change argument identifies it with risk-neutral pricing under $\mathbb{Q}$.

The European Call Formula

The call payoff splits into two truncated expectations

Let

$$g(s) = (s - K)^+, \qquad \tau = T - t.$$

Under the auxiliary drift-r dynamics,

$$S_T^r = s \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\right).$$

Therefore

$$C(t,s) = e^{-r\tau}\left(\mathbb{E}_{t,s}^r[S_T^r \mathbf{1}_{\{S_T^r > K\}}] - K\mathbb{P}_{t,s}^r(S_T^r > K)\right).$$

The exercise event creates d_-

YOUR TURN Rewrite $S_T^r > K$ as a threshold event for $Z \sim N(0,1)$.

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YOUR TURN Rewrite $S_T^r > K$ as a threshold event for $Z \sim N(0,1)$.

$$S_T^r > K \iff Z > \frac{\log(K/s) - (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}} = -d_-,$$

where

$$\boxed{d_- = \frac{\log(s/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.}$$

Thus

$$\boxed{\mathbb{P}^r_{t,s}(S_T^r > K) = \mathrm{N}(d_-).}$$

A stock-weighted exercise probability shifts the Gaussian

The remaining call term is $\mathbb{E}^r_{t,s}[e^{aZ}\mathbf{1}_{\{Z>-d_-\}}]$, where $a = \sigma\sqrt{\tau}$. Unlike the strike term, it weights exercise states by the stock price and therefore favors larger Z.

YOUR TURN Complete the square to evaluate this weighted Gaussian tail.

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YOUR TURN Complete the square to evaluate this weighted Gaussian tail.

Letting φ be the standard gaussian density.

$$e^{az}\varphi(z) = e^{a^2/2}\varphi(z-a),$$
$$\mathbb{E}^r_{t,s}[e^{aZ}\mathbf{1}_{\{Z>-d_-\}}] = e^{a^2/2}\mathrm{N}(d_- + a).$$

Thus $\boxed{d_+ = d_- + a}$: d_- measures exercise probability, while d_+ prices the stock delivered in those exercise states.

The Black–Scholes call formula follows

The stock part becomes

$$\mathbb{E}^r_{t,s}[S^r_T \mathbf{1}_{\{S^r_T > K\}}] = s e^{r\tau} \mathrm{N}(d_+).$$

Hence

$$\boxed{C(t,s) = s\mathrm{N}(d_+) - K e^{-r\tau} \mathrm{N}(d_-),}$$

with

$$\boxed{d_\pm = \frac{\log(s/K) + r\tau}{\sigma\sqrt{\tau}} \pm \frac{1}{2}\sigma\sqrt{\tau}.}$$

SIMPLIFICATION This closed form uses constant r, σ, no dividends, European exercise, and lognormal stock dynamics.

The two normal terms have different meanings

$$C(t,s) = \underbrace{s\mathrm{N}(d_+)}_{\text{stock delivered in exercise states}} - \underbrace{Ke^{-r\tau}\mathrm{N}(d_-)}_{\text{discounted strike paid}}.$$

Probability warning

$\mathrm{N}(d_-)$ is the exercise probability under the auxiliary pricing dynamics. $\mathrm{N}(d_+)$ is not the same probability; it arises from a stock-weighted expectation.

Useful relation

$$d_+ - d_- = \sigma\sqrt{\tau}.$$

The formula survives basic boundary checks

YOUR TURN Predict the limits when $s \downarrow 0$, $s \uparrow \infty$, and $\tau \downarrow 0$.

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$$
\begin{array}{lll}
s \downarrow 0: & d_{\pm} \to -\infty, & C(t,s) \to 0, \\
s \uparrow \infty: & d_{\pm} \to +\infty, & C(t,s) \sim s - Ke^{-r\tau}, \\
\tau \downarrow 0: & & C(t,s) \to (s-K)^{+}.
\end{array}
$$

Also $0 \le C(t,s) \le s$.

Why check limits?

They catch sign mistakes in discounting, $\tau = T - t$, and $d_{\pm}$.

Put–call parity gives the put without another integral

At maturity,

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K.$$

Replication and no arbitrage imply

$$\boxed{C(t,s) - P(t,s) = s - Ke^{-r\tau}.}$$

YOUR TURN Deduce the Black–Scholes put price.

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YOUR TURN Deduce the Black–Scholes put price.

$$\boxed{P(t,s) = Ke^{-r\tau}\mathrm{N}(-d_-) - s\mathrm{N}(-d_+).}$$

Greeks Are Hedge and Model Diagnostics

One density identity makes every derivative shorter

Let

$$C = s\mathrm{N}(d_+) - Ke^{-r\tau}\mathrm{N}(d_-), \qquad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

YOUR TURN Show that the two density terms satisfy a simple identity.

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YOUR TURN Show that the two density terms satisfy a simple identity.

Using $d_- = d_+ - \sigma\sqrt{\tau}$,

$$\frac{\varphi(d_-)}{\varphi(d_+)} = \exp\left(\frac{d_+^2 - d_-^2}{2}\right) = \frac{se^{r\tau}}{K}.$$

Thus

$$\boxed{s\varphi(d_+) = Ke^{-r\tau}\varphi(d_-).}$$

Delta is the stock holding in the replicating strategy

$$C(t,s) = s\mathrm{N}(d_+) - Ke^{-r\tau}\mathrm{N}(d_-), \qquad d_\pm = \frac{\log(s/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

YOUR TURN Starting from the displayed formula, differentiate $C(t,s)$ with respect to s. Keep both $d_\pm$ terms until they cancel.

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YOUR TURN Starting from the displayed formula, differentiate $C(t,s)$ with respect to s. Keep both $d_\pm$ terms until they cancel.

Since

$$\partial_s d_+ = \partial_s d_- = \frac{1}{s\sigma\sqrt{\tau}},$$

$$\Delta = C_s = \mathrm{N}(d_+) + s\varphi(d_+)\partial_s d_+ - Ke^{-r\tau}\varphi(d_-)\partial_s d_-.$$

The density identity gives

$$\boxed{\Delta_{\text{call}} = \mathrm{N}(d_+).}$$

Gamma measures how fast the hedge must move

YOUR TURN Differentiate $\Delta = \mathrm{N}(d_+)$ once more with respect to s.

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$$\Gamma = C_{ss} = \varphi(d_+)\partial_s d_+ = \boxed{\frac{\varphi(d_+)}{s\sigma\sqrt{\tau}}.}$$

Interpretation

$\Gamma > 0$ for a call. The hedge changes most rapidly near the strike and near maturity.

MARKET ASSUMPTION Continuous rebalancing is costless in the model. In practice, large gamma means frequent trading, costs, and discrete-hedging error.

Vega quantifies volatility-model sensitivity

YOUR TURN Differentiate the call price with respect to σ. Use the density identity instead of expanding every term separately.

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The $d_{\pm}$-derivative terms simplify to

$$\boxed{\text{Vega} = C_\sigma = s\varphi(d_+)\sqrt{\tau}.}$$

Interpretation

Vega is positive: more volatility increases the value of a convex payoff.

SIMPLIFICATION Black–Scholes treats σ as known and constant. Market implied volatility varies by strike and maturity.

Theta is quickest to compute from the PDE

YOUR TURN Use the PDE with the known call delta and gamma to find $\Theta = C_t$.

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$$\Theta = rC - rs\Delta - \frac{1}{2}\sigma^2 s^2 \Gamma.$$

Substitution gives

$$\boxed{\Theta_{\text{call}} = -\frac{s\sigma\varphi(d_+)}{2\sqrt{\tau}} - rKe^{-r\tau}\mathrm{N}(d_-).}$$

Convention

Here theta means $\partial C/\partial t$, calendar time moving forward. Some systems report the opposite sign or a one-day rescaling.

Rho measures sensitivity to the financing rate

YOUR TURN Differentiate $C(t, s)$ with respect to r.

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YOUR TURN Differentiate $C(t, s)$ with respect to r.

Because

$$\partial_r d_+ = \partial_r d_- = \frac{\sqrt{\tau}}{\sigma},$$

the density contributions cancel, leaving

$$\boxed{\mathrm{Rho}_{\mathrm{call}} = C_r = K\tau e^{-r\tau}\mathrm{N}(d_-).}$$

SIMPLIFICATION The formula assumes a single deterministic rate used for discounting, borrowing, and lending. Real funding and collateral conventions are richer.

Put Greeks follow from parity

Differentiate

$$P = C - s + Ke^{-r\tau}.$$

YOUR TURN Find the put delta, gamma, vega, theta, and rho.

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$$\begin{aligned}
\Delta_P &= \mathrm{N}(d_+) - 1 = -\mathrm{N}(-d_+), \\
\Gamma_P &= \Gamma_C, \qquad \mathrm{Vega}_P = \mathrm{Vega}_C, \\
\Theta_P &= -\frac{s\sigma\varphi(d_+)}{2\sqrt{\tau}} + rKe^{-r\tau}\mathrm{N}(-d_-), \\
\mathrm{Rho}_P &= -K\tau e^{-r\tau}\mathrm{N}(-d_-).
\end{aligned}$$

The five main Greeks answer different questions

Greek	Formula for a call	Main risk question
Δ	$\mathrm{N}(d_+)$	How many shares hedge?
Γ	$\varphi(d_+)/(s\sigma\sqrt{\tau})$	How fast does delta change?
Vega	$s\varphi(d_+)\sqrt{\tau}$	What if volatility is wrong?
Θ	$-s\sigma\varphi(d_+)/(2\sqrt{\tau}) - rKe^{-r\tau}\mathrm{N}(d_-)$	What does time cost?
Rho	$K\tau e^{-r\tau}\mathrm{N}(d_-)$	What if rates change?

Hierarchy

Delta is the model hedge. Gamma controls hedge instability. Vega, theta, and rho expose parameter and carry sensitivity.

A Greek is local, while hedging error accumulates

With discrete trading over Δt,

$$\Delta C \approx \Delta\, \Delta S + \Theta\, \Delta t + \frac{1}{2}\Gamma(\Delta S)^2.$$

YOUR TURN Which idealizations become most fragile for a high-gamma option?

A Greek is local, while hedging error accumulates

With discrete trading over Δt,

$$\Delta C \approx \Delta\, \Delta S + \Theta\, \Delta t + \frac{1}{2}\Gamma(\Delta S)^2.$$

YOUR TURN Which idealizations become most fragile for a high-gamma option?

- continuous trading;
- zero transaction costs and unlimited liquidity;
- continuous stock paths;
- known constant volatility.

What the Derivation Really Established

The complete chain is one reproducible argument

GBM model + frictionless complete market

$$\Downarrow$$

self-financing replication with $\Delta = p_s$

$$\Downarrow$$

$$p_t + rsp_s + \tfrac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0$$

$$\Downarrow$$

$$p(t,s) = e^{-r\tau}\mathbb{E}[g(S_T^r) \mid S_t^r = s]$$

$$\Downarrow$$

$$C = s\mathrm{N}(d_+) - Ke^{-r\tau}\mathrm{N}(d_-).$$

Final check: identify the decisive assumption

YOUR TURN

1. The $\mathrm{d}W$ exposure can be eliminated.
2. The hedged position earns r.
3. The call has a $d_{\pm}$ closed form.
4. Delta can be followed exactly.

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1. One traded stock spans the one Brownian risk: completeness.
2. No arbitrage and unrestricted borrowing/lending at r.
3. Constant r, σ, lognormal dynamics, no dividends, European payoff.
4. Continuous frictionless trading with predictable admissible strategies.

Keep these six ideas

1. Itô's second-order term creates gamma exposure.
2. Self-financing is a gain equation, not merely a value identity.
3. Matching diffusion determines delta.
4. Matching drift gives the PDE and removes μ.
5. The logarithmic transform turns the PDE into heat flow.
6. Verification connects the PDE solution, discounted expectation, and replicating strategy.

Next conceptual step

Risk-neutral measure changes explain probabilistically why the auxiliary drift is r.

Optional extension: continuous dividends

Suppose the stock pays a continuous dividend yield q. Under the risk-neutral measure,

$$\frac{\mathrm{d}S_t}{S_t} = (r - q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t.$$

For a European call with payoff $(S_T - K)^+$,

$$C(t, s) = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+ \mid S_t = s\right], \qquad \tau = T - t.$$

YOUR TURN Derive the pricing PDE and the call formula. What changes in the stock term, and what stays the same in the strike term?